



NUS
National University
of Singapore



Team LeoNUS – CanSat Competition Team

Engineering the Future, One CanSat at a Time

AAS CanSat Competition

- Annual aerospace challenge by AAS and NASA
- 80+ university teams from 20–30 countries
- Design & launch a satellite the size of a soda can
- Launched to ~1 km, performing real mission tasks
- Hands-on gateway to aerospace engineering



NUS
National University
of Singapore



CANSAT COMPETITION



Get a taste for your future in space — Enter today!

The CanSat Competition is an annual event where college and university student teams from around the world compete in a challenge to design, build, and launch a payload system. Teams are scored based on their ability to fulfill mission requirements.

Hosted by the American Astronautical Society and the Naval Research Lab, the CanSat Competition is recognized internationally as one of the best tests of an aerospace education - a notable addition to any resume.

How to Compete:

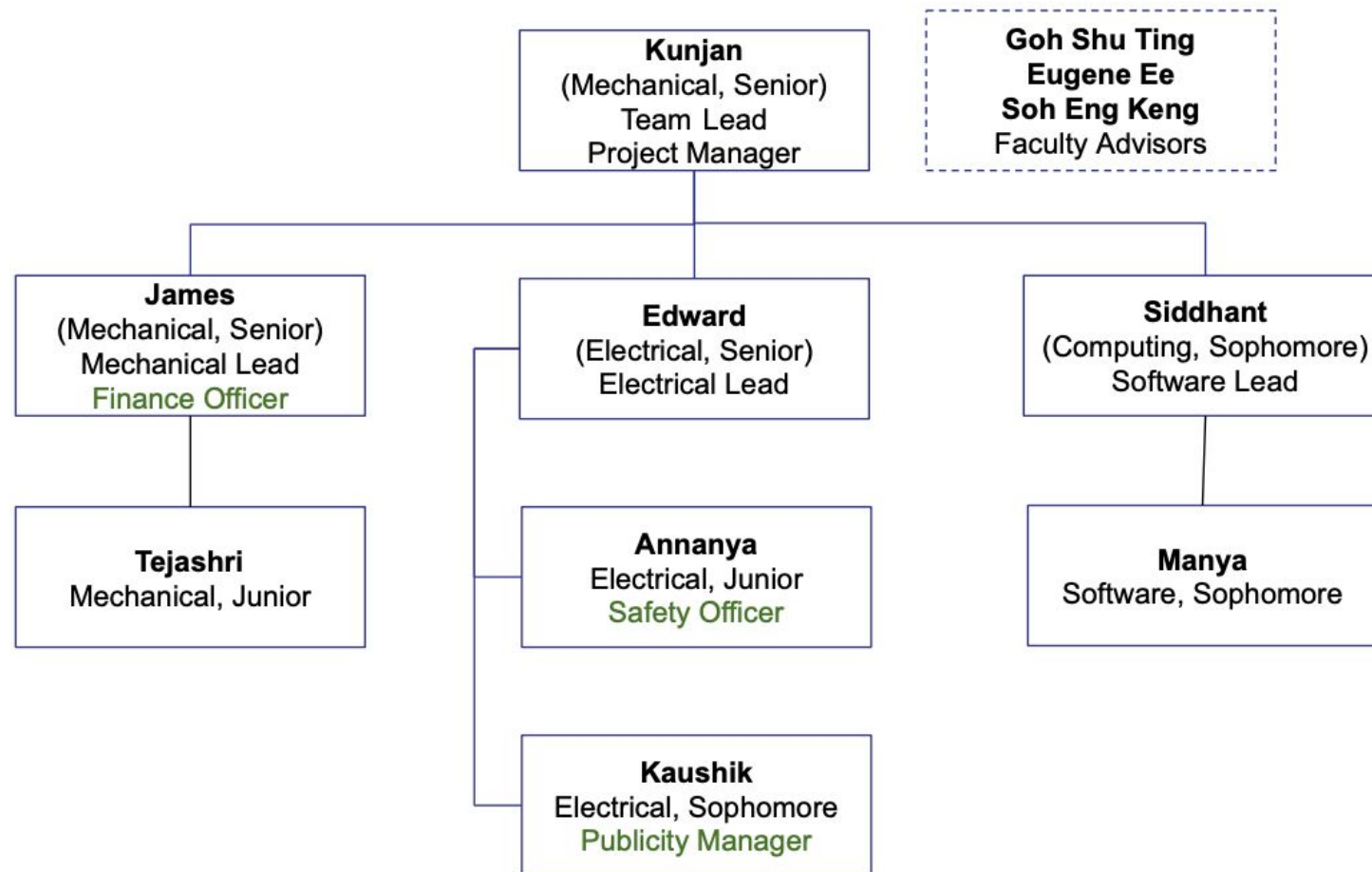
- **Assemble Your Team:** Teams can be undergraduate, graduate, or a mix.
- **Secure a Faculty Advisor:** Teams need a school staff member to help advise and guide them through the competition.
- **Apply and Compete:** Submit your team application by November 4, 2023 to be considered for eligibility.

Register your team by Nov. 4 on
cansatcompetition.com



Meet Team LeoNUS

8 dedicated undergraduate members: Our core team comprises interdisciplinary students from Mechanical, Electrical, and Computer Engineering, many from the innovative iDP programme.



Team LeoNUS kick-off meeting



Mechanical Engineering

Designing robust structures and deployment mechanisms.



Electrical Engineering

Developing power systems and sensor integration.



Computer Engineering

Programming flight software and data acquisition.

Our Track Record: Consistent Global Performance

2024: Strong Debut

Qualified & achieved a **9th place internationally** with a team of 9 final-year project (FYP) students.

1

2

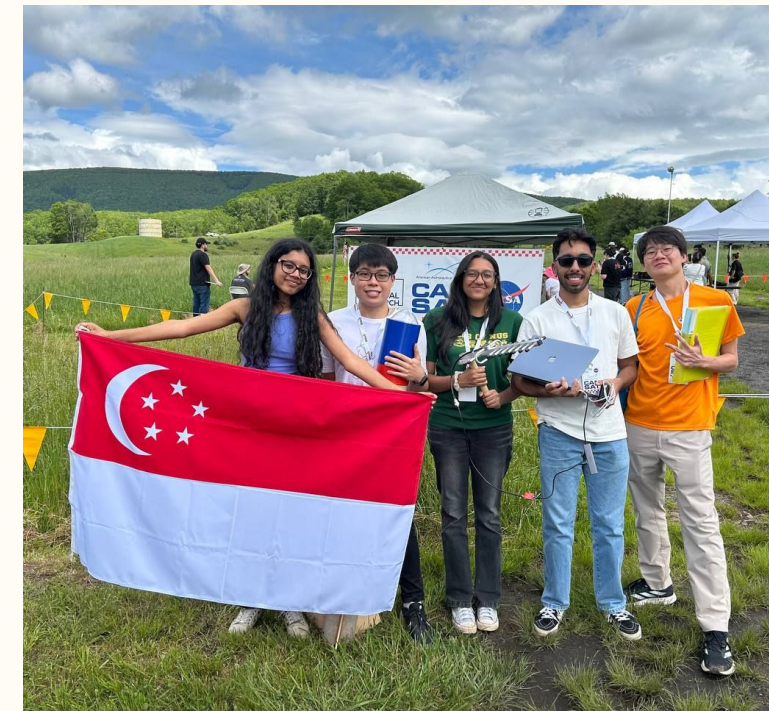
3

2026: Building Momentum

Currently preparing for our **third consecutive campaign**, integrating FYP, Undergraduate Research Experience (UREX), and returning members to leverage accumulated knowledge.

2025: Sustained Success

Qualified again, placing **24th globally**. This achievement was particularly significant as the team was sustained by a small number of students with interest in the space programme.



Vision for the Future: 2026-2027 Strategic Roadmap

Team LeoNUS is not just participating; we are strategically planning for sustained growth and global leadership in the CanSat arena.

1

2026: Elevating Performance

Target Top 5: Aiming for a top-five finish in the international competition.

Ground Station Award: Striving for excellence in data reception and analysis.

Subsystem Development: Innovating new CanSat components for enhanced mission capabilities.



2

2027: Sustainable Leadership

Team Expansion: Integrating FYPs from diverse faculties to broaden interdisciplinary impact.

Formalization: Becoming a flagship team under EDIC, solidifying our institutional standing.

Sustainable Pipeline: Establishing a robust pipeline for future NUS aerospace talent.



Current Support Landscape

Facilities and Support by iDP and ECE

- **EDIC Workshop Facilities**
- **Innovation and Design Hub**
 - Prototyping and Fabrication facilities(3D printing, laser cutting etc.)
- **STAR(Satellite Technology and Research Centre)**
 - Thermal Chamber
 - Vacuum Chamber
 - Vibration Testing Device

Current Funding

- **iDP FYP Budget: S\$800 × 3 students = S\$2,400**
Supports prototyping and fabrication of the CanSat system.
- **UREX Grant: S\$5,000**
Allocated primarily to competition logistics for the 2026 international campaign.



Funding Support Request

Additional Budget Requirement

→ **S\$60,000 – S\$65,000 (AY25-26)**

Key Funding Areas:

 **Travel & Logistics:** Flights, accommodation, and equipment freight for 10 students.

 **Testing & Equipment:** Drop-test system and ground-station device.

 **Prototyping & Materials:** Sensors, fabrication materials, and electronics.

 **Outreach & Training:** Workshops, exhibitions, and student engagement.

Budget Breakdown

Category	Estimated Cost (S\$)	Details
Travel (10 pax, 6 days)	~45,000	Flights, accommodation, daily allowance
Visa & Insurance	2,000	-
Drop-Test Equipment	11,000	Balloon/drone test system and safety accessories
Outreach & Merchandise	1,500	Showcase Material and Team Uniforms
Ground-Station Device	2000	-
Total	60,000-65,000	

The urgent need for Deanery funding is paramount to sustain our operations, expand our reach, and enable us to truly compete on a global scale.

The background is a blue-toned illustration. At the top center, a rocket is launching upwards, leaving a trail of smoke. To the right, the Indonesian flag (Merah Putih) is waving. In the foreground, several people are shown from behind, looking up and reaching their hands towards the sky in a gesture of celebration or aspiration. The overall scene conveys a sense of achievement and future-oriented vision.

Thank You

We're not just building satellites, we're
building the next generation of aerospace
innovators.